

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $f, g \in L^1(X)$. I'm trying to show that the set:

$$\{x \mid |f(x)| \leq g(x)\}$$

is a measurable set. The above set can be written as:

$$\{x \mid 0 \leq f(x) + g(x)\} \cap \{x \mid 0 \leq g(x) - f(x)\}$$